

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Monitored production systems and debugged tenant-specific incidents; hardened auth and rate limiting, added **AWS KMS** encryption for PII/clinical data, and remediated **20+ VAPT findings**.
- Served as primary technical POC for the eTriage backend, ran knowledge transfer onboarding a second developer to on-call readiness in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Shipped a production LLM feature running a **4-bit quantized Llama-7B via llama.cpp** on a single VM under constrained compute, with async queued generation, per-persona prompts, and content-safety filtering.
- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation and clear contracts, and migrated core repositories from JavaScript to TypeScript.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN gym-management platform with Stripe + manual-cash hybrid payments, cron automation, and multi-branch support; live **3+ years**.

Projects

Sonare – Offline On-Device Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop app with a fully **on-device inference pipeline** (MediaPipe hand tracking, gesture classification, whisper.cpp ASR, local TTS) running offline with no network dependencies; designed async APIs and queues to keep the UI responsive under real-time video and audio streams.

Refusal Decay – LLM Safety / Mechanistic Interpretability (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Probed the internals of Llama-3.1-8B-Instruct, extracting the residual-stream refusal direction via difference-in-means (refusal **0.92** → **0.32** at $k=3$) with causal activation patching through PyTorch forward hooks, bootstrap 95% CIs, and McNemar's test.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** dataset via broadcast joins, AQE, and skew-join handling, using Spark UI stage timings to localize shuffle and skew bottlenecks and reduce the tail-latency ratio **2.14x to 1.74x**.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, C++, TypeScript, JavaScript, Java, SQL, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: FastAPI, Node.js, Express, Docker, Git, GitHub Actions, pytest, REST APIs

ML & AI: PyTorch, Transformers, model quantization (llama.cpp 4-bit), edge / on-device inference, JAX, NumPy

Systems / Performance: Spark / PySpark, Spark UI profiling, latency & memory analysis, cache-aware design, Linux perf

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.